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Hybrid control paradigm for exploring VR teleoperation and DRL-driven autonomy in mobile robotics

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Abstract

Recent studies have suggested ways to enhance user perception of remote workspaces and robot autonomy, often at the expense of system suitability and efficiency. This research introduces a novel method, Distributed Supervisory Control (DSC), leveraging Virtual Reality (VR) to enhance teleoperation. The DSC method intelligently distributes tasks between the robot and the human operator, minimizing shared autonomy conflicts and sensory data transfer. It uses Deep Reinforcement Learning (DRL) through a Twin Delayed Deep Deterministic Policy Gradient (TD3) method for tasks like obstacle avoidance. Real-time experimentation validated the method's effectiveness through performance metrics, including NASA-TLX, task execution time, and obstacle collision frequency. Kruskal-Wallis tests identified significant differences in task execution times and collision frequency across DSC, Direct Control (DC), and Assistive Direct Control (ADC). Dunn's post-hoc tests indicated DSC significantly outperformed DC and ADC in both execution time and collision frequency. Similarly, NASA-TLX scores for effort, mental demand, performance, frustration, physical demand, and temporal demand also showed significant differences ($p < 0.05$), supporting DSC's lower task load. User case studies revealed enhanced user experience, as measured by the System Usability Scale (SUS) and Igroup Presence (IPQ) questionnaires for immersive experience. The non-parametric Kruskal-Wallis test was utilized to verify significant differences between group medians, followed by a Dunn's test that revealed significant performance improvements with our Distributed Supervisory Control (DSC) method relative to Direct Control (DC) and Assistive Direct Control (ADC), thereby enhancing task efficiency and overall interface suitability. In conclusion, Distributed Supervisory Control significantly enhances task efficiency and user experience in robot teleoperation environments, demonstrating its potential as a new standard for remote operations.

Keywords Virtual reality · Teleoperation · Distributed supervisory control (DSC) · Actor-Critic network · Deep reinforcemnt learning (DRL)

Nomenclature

P_{ref} Reference position passed from the virtual environment.

Extended author information available on the last page of the article

P_t Current Robot Pose data calculated through odometry.
ε Tolerance between set and reached position.
d_g Robot's initial distance from the goal point.
α Robot's initial heading towards the goal point.
θ_g Relative angle between the robots heading and the heading towards the goal.

1 Introduction

Recent advancements in robotic technology, artificial intelligence, and computer vision have broadened the spectrum for the adoption of mobile robots. These mobile agents are being utilized for extraction tasks [1–3], space exploration [4, 5], for defense purposes [6], and industrial uses [7, 8]. This adoption is expected to surge even more in the coming decades. With around 3 million industrial robots globally, the International Federation of Robotics reports a 37% growth in professional service robots and a 19% increase in domestic service robots over the past year, indicating growing interest in robotics [9]. However, robotic applications have potential for further advancement by enhancing reliability and emulating human intelligence through self-reliant behavior. The objective of these applications of attaining full autonomy is still elusive, as even the more advanced robots still require human intervention due to the complex nature of real-world environments [10]. This complexity aggravates even further when these operating environments turn remote, making the requirement of external assistance inevitable [11].

However, this predicament can be managed through robotic teleoperation. Teleoperation significantly enhances remote capabilities by allowing humans to safely manage tasks in hazardous or inaccessible environments. Despite this advantage, current teleoperation systems still have limitations, particularly limited operator situational awareness, cognitive overload, and high communication latency resulting from computational complexity and extensive data transfer. The teleoperation process involves transmitting supervisory commands from the operator via a communication network to a remote robot, while simultaneously receiving feedback about the robot's state and the environment. By effectively transmitting and receiving this information, these teleoperated robotic systems have proven favorable for tackling harsh operating environments. Such environments include areas of high radioactivity [12, 13], aerial spaces [14, 15], and in water bodies [16, 17]. The ability to operate in these challenging conditions highlights the robustness of teleoperated systems. Nevertheless, the true success of these systems hinges on the operator's capability to provide efficient control commands. This capability directly depends on the accuracy and intuitiveness of the operator's perception and understanding of the remote environment [18]. Insufficient perception significantly reduces operators' decision-making effectiveness and responsiveness in complex teleoperation tasks.

To overcome these limitations, researchers generally adopt two distinct approaches. The first involves shared autonomy, a control paradigm where the human operator and the robot work in tandem to decide the best course of action during the operation, thus reducing the dependence on user decisiveness and skills [11]. This approach, however, raises concerns, including potential conflict between human and robotic decision-making, leading to operator discomfort [19] and the additional computational requirement for comparing user-robot choices [11]. To counter these issues, our study proposes a Distributed Supervisory Control

(DSC) method. This system enables the human operator to retain control over higher-order tasks, such as determining the destination waypoints. Simultaneously, the robot is endowed with autonomy to oversee lower-order tasks that demand instinctual responses, like avoiding obstacles while plotting a trajectory to reach the specified waypoints and eventually, the destination. This beneficial division of tasks contributes to system parallelism described in Sect. 3, which subsequently mitigates the additional computational demand [11].

The second approach aims to improve the operator's understanding of the remote environment by increasing the amount of sensory feedback. While this approach does enhance the operator's perception, it also raises the complexity and computational demands for data transmission and interpretation. To alleviate this, our work focuses on designing an intuitive and immersive interface. This interface mitigates the reduced information feed without compromising the operator's perception. In the present study, the operator's contextual understanding of the remote environment is constructed dynamically based solely on sensor data streamed back from the robot during exploration, rather than from any pre-existing map or model. This ensures that the operator's view reflects only areas that have been directly sensed and reported by the robot in real time. As a result, we intend to establish a foundation for more autonomous, yet human-supervised robotic operations in challenging environments, effectively addressing current teleoperation limitations.

2 Related research

Cutting-edge research proposes two major strategies to ameliorate teleoperation [20]. The first involves less dependency on operator ability due to enhanced robot autonomy through a shared control approach. The second strategy improves the operator's perception of the remote environment. This enhanced perception leads to better situational awareness [11]. The approaches that concentrate on improving the operator's awareness of the distant locale include the augmentation of the user viewpoint [20], tactile consciousness [21] and virtual presence [22]. The idea behind these is to help create a user experience that forms a deep connection with the far-off environment, escalating their situational comprehension, and consequently boosting the accuracy of the entire procedure. However, these approaches involve exchanging large amounts of information. Processing such information requires significant computational resources [23]. This bulk information exchange inherits a transfer lag that may negate the positive gain it produces [11]. Our research addresses teleoperation challenges by developing an intuitive and immersive interface that enhances operator comprehension while reducing data transfer and computational demands through onboard data compression and compact information packets.

An array of investigations has been put forward about shared autonomy. These include deploying human-in-loop reinforcement learning policies for improved task completion percentages [24], illustrating navigation path sketches [25], and some have involved probabilistic forecasting of the target points [26]. In all these approaches, the robot possesses local intelligence to complete the action loop. However, this autonomy increases the risk of disagreements between robot decisions and operator intentions. Research has highlighted that this intentional mismatch can lead to delayed decision-making [27]. A remedy for preventing this includes revealing the robot's objectives to the operator in advance [28]. Nevertheless, the research that evaluated operators' feedback on actions proposed by a robot

discloses that the operators tend to override the actions suggested by the robots [29]. Thus, nullifies the proposed remedy. To resolve this [30], presented an illusory control method that conceals the robot operation from the operator during potential intention discrepancies, such as evading an obstacle, meanwhile allowing the operator to control a virtual robot. However, this approach adds to the costs by necessitating the creation of visual illusions [20], and at the same time, diminishes the operator's supervision that could be beneficial for the robot in cases when autonomous navigation becomes problematic. Thereby, our proposed work evolves a distributed task control methodology where the operator possesses the capability to broaden their role by providing the option to the user to intervene, if necessary.

Communication lags have always been a concern with teleoperated systems. Recent studies have proposed predictive displays as a solution for this enduring sore point [31–35]. These methods involve the use of image transformation algorithms to calculate and visually represent future robot positions for the operator [33], aiming to support the operator in making behavioral alterations based on these future predictions. Contrastingly, our proposed methodology enhances a robot's local autonomy for executing navigational tasks, thereby decreasing reliance on operator feedback that may potentially be influenced by communication delays, as detailed in Sect. 4.3. This aspect of local autonomy often finds manifestation in autonomous obstacle avoidance procedures [36, 37].

Parallel to this, existing research has proposed map-dependent autonomous navigation schemes [38]; however, such methodologies can quickly spiral into complexity in the face of dynamic and complex environments [39], necessitating additional resources for prior environment mapping. Recently, Deep Reinforcement Learning (DRL) has emerged as a promising approach for robot navigation tasks. These techniques enable accurate decision-making, especially in unknown and dynamic environments [40]. Recent works, for instance [41] proposed to leverage deep reinforcement learning methods, specifically employing Quality-Diversity-based Reinforcement learning (RL) techniques to improve teleoperation performance through real-time visualization and semi-autonomous control. However, these approaches inherently involve high computational complexity and intensive real-time data communication, with limitations in simulation-to-real-world transferability. Thereupon, our work centers on the application of a goal-driven Twin Delayed Deep Deterministic Policy Gradient approach, designed specifically for directional local navigation tasks. This method efficiently operates with sparse environmental state data. This results in reduced computational overhead and enhanced sim-to-real transferability, making it highly suitable for practical real-world deployments. Moreover, in this study, the operator's understanding of the remote unknown environment is developed in real time based exclusively on sensor data streamed back from the robot during exploration, rather than from any pre-existing 3D model or static map. For instance, a typical realistic use scenario in a search-and-rescue real world scenario, a mobile robot deployed to a collapsed building sends live sensor data as it uncovers new spaces, and the operator's VR view accommodates the perspective of the robot in the real world.

Simultaneously, the emergence of Virtual Reality (VR) technology in the industrial landscape is revolutionizing robot teleoperation, primarily due to its immersive capacities that enhance operator situational awareness, intuitive spatial comprehension, and overall ergonomic comfort [11]. While alternative non-VR methods exist, VR provides distinctive advantages when quick interpretation of spatial information is necessary. The inherent immersive nature of VR expedites situational awareness and fosters a more intuitive control

interface, attributes that are indispensable in high-precision, high-dexterity applications [42, 43]. Notable recent work [44], demonstrated improvements in teleoperation performance, operator intuitiveness, and ergonomic benefits through a VR-based immersive teleoperation system, underscoring its potential to enhance operator efficiency and reduce physical fatigue. However, such immersive interfaces inherently demand significant computational resources and high communication bandwidth for maintaining seamless real-time responsiveness. Conventionally, these interfaces rest on the use of pre-constructed 3D models of operational spaces [30]. This approach is particularly beneficial for VR-based teleoperation of robotic manipulators, which usually function within a stable work environment, allowing for efficient and effective use of these static 3D models [45, 46]. In contrast, mobile robots, which are designed to function within dynamic and unpredictable environments, present a unique set of challenges. Mobile robots typically require real-time updates of their 3D remote environments. Providing these updates involves considerable complexity and significant computational resources [47]. To overcome this, our focus is on formulating virtual environments with fewer components that are vital while retaining depth perception and meaningful information representation. Existing research underscores our approach, indicating that operators' depth perception and spatial understanding can significantly improve, even within simplified VR environments [48, 49]. Considering the ergonomic requirements [50], our focus extends to the ergonomics of the operation interface by developing a standalone application that minimizes wire handling issues. These ergonomic improvements are crucial as they directly impact operator efficiency, comfort, task performance, and fatigue reduction [51, 52].

Taking this ergonomic focus into consideration, integrating Virtual Reality (VR) technology into our teleoperation serves a dual purpose [53, 54]. First, VR enhances system capabilities by offering an interactive environment for better control and precision, while also enriching user experiences through realistic simulations and intuitive interfaces, thus reducing cognitive load. Second, VR supports ergonomic improvements by enabling more natural and comfortable operator interactions, reducing physical strain and fatigue. Collectively, these improvements not only optimize system performance but also enhance operator engagement, making our approach a well-rounded solution for advanced teleoperation tasks.

To identify existing gaps in recent robot teleoperation, VR interfaces, and reinforcement learning approaches, Table 1 provides a concise comparison of relevant methodologies. The comparative summary highlights methods, interfaces, autonomy levels, and their limitations, thus explicitly validating improvements introduced by our proposed method described subsequently.

In summary, the persistence of these challenges from operator workload and autonomy conflicts to computational, perceptual, and communication barriers indicates that a comprehensive solution remains an open need in the field. The Distributed Supervisory Control (DSC) method that we present in this work is formulated directly in response to these interrelated issues, with each component corresponding to the key gaps described above. The way in which our approach addresses these challenges is shown in Table 1 and further explained in Sect. 2.1.

Table 1 Comparative summary of related methods and their limitations

Reference	Methodology	Operator Interface	Autonomy Level	Communication and Data Management	Identified Limitations	Improvements in Proposed Method
[20]	Illusory VR Control	VR with visual illusions, pre-constructed 3D models	Shared autonomy with virtual guidance	Visual illusions, high bandwidth required	High computational load, reduced operator supervision	Lower computational load; intuitive supervision via simplified VR.
[25]	Human-in-Loop Reinforcement Learning	Non-VR standard interface	Shared autonomy (robots propose policies, operators can override)	Standard (no special data management)	Operator-robot disagreement conflicts; overrides cause delays	Conflict-free, seamless user override integration.
[33]	Predictive visual feedback	Predictive visual/non-VR Interface	Semi-autonomous with operator influencing predictions	Predictive modeling, latency-sensitive	Computational overhead, latency vulnerability	Reduced predictive complexity and latency sensitivity.
[38]	Map-dependent autonomous navigation	Non-VR interface, detailed environmental mapping required	Fully autonomous navigation	Requires extensive pre-mapping data exchange	Inefficient for changing/dynamic environments, high computational load.	No extensive pre-mapping; efficient real-time sparse-environment DRL autonomy.
[40]	Deep reinforcement learning (DRL)-based autonomous navigation	Non-VR regular interface, deep RL-based navigation	Autonomous navigation in unknown environments	No explicit data-reduction or VR considerations	High computational demand, potential sensory overload for operators if converted to VR contexts.	Optimized VR interface; minimal computational overhead.
[41]	DRL-based non-VR teleoperation	Non-VR GUI, real-time action visualization	Mixed autonomy	High real-time communication, computational load	High complexity, data load, and limited sim-to-real transferability.	Lower computational overhead; High sim-to-real transferability
[44]	Immersive VR teleoperation with ergonomic controls	Ergonomic VR interface (glove, joystick)	Operator-controlled; minimal autonomy	Ergonomic design without optimized data transmission strategies	High data bandwidth requirements with complex interfaces.	Ergonomic interface design coupled with efficient, reduced-data exchange.

2.1 Contributions

Motivated by the limitations identified in existing methods Table 1, this study introduces a novel Distributed Supervised Control (DSC) teleoperation method, enhanced by deep reinforcement learning (DRL) driven local autonomy. This approach addresses the shortcomings of existing teleoperation methods while integrating their strengths. Specifically, the research proposes a control methodology that achieves three primary objectives: (1) reducing the operator's workload, (2) enhancing task efficiency, and (3) improving overall system suitability, including providing more immersive user experience.

In developing the DSC method, we implemented a unique approach to parallel supervisory control. This approach allows human operators to handle higher-order tasks like setting destination waypoints, while robots autonomously manage lower-order tasks such as obstacle avoidance and trajectory plotting. Additionally, our system integrates visual feedback in a virtual environment, capturing only essential spatial features like obstacle depth and vital robot information. This minimizes the computational load and reduces the costs and processing power required for a full 3D environment, especially as the robot explores new areas [30].

A key innovation in this study is the deployment of a deep reinforcement learning (DRL) framework trained to operate with sparse environmental state information, more specifically a goal driven Twin Delayed deep deterministic approach that enables the robot to autonomously navigate and manage both static and dynamic obstacles, such as unexpected human or object intrusions. By leveraging this DRL framework, the robot can adapt to varying scenarios in real-time without relying heavily on continuous input from the human operator. As a result, the system significantly reduces the human operator's workload, allowing them to focus on higher-level decision-making tasks, and enhances overall task efficiency, thereby improving the effectiveness of remote robotic control in dynamic environments.

The research enhances operator comprehension through an intuitive interface while minimizing computational complexities in data transmission. By employing onboard data compression and efficient messaging protocols, we effectively reduce data volume, latency, and computational load. Additionally, we introduce a technique for generating a 2D point cloud overlay in the virtual environment, utilizing advanced line rendering to delineate clear boundaries from sparse planar points. This combination of efficient data handling and precise visual representation optimizes operator performance and ensures system responsiveness in complex scenarios. To summarize, this research includes:

- Introduction of the Distributed Supervised Control (DSC) method, which reduces operator workload by automating lower-order tasks through reinforcement learning.
- Enhancement of robot navigation in dynamic environments using a goal-driven Twin Delayed Deep Deterministic policy, boosting task efficiency and effectiveness.
- Optimization of visual feedback and data handling techniques, improving overall system suitability and responsiveness.

This study explicitly aims at enhancing single-robot teleoperation scenarios using VR by addressing well-established limitations, such as constrained operator situational awareness, operator workload, and latency-related control issues. The proposed Distributed Supervisory Control (DSC) method directly and explicitly addresses these challenges through:

- Distributed control between waypoint-based human supervision (high-level interaction), and local autonomous navigation via goal-driven Deep Reinforcement Learning (TD3), significantly reducing latency sensitivity and operator workload.
- Utilization of intuitive, simplified Virtual Reality interfaces, explicitly enhancing operator awareness without overwhelming sensory bandwidth.

The remaining part of the paper is constructed as follows: Sect. 3 presents the DSC method, while Sect. 4 presents the user case studies elaborating the method's operability, and Sect. 5 concludes the paper with the help of findings from Sect. 4.

3 Methodology

3.1 System brief

The conceptual flow of the DSC method is described by Fig. 1. This method enables parallel control of a remote mobile robot by utilizing a virtual twin. The core operations of the DSC system can be summarized as follows:

- *Perception Building*: The system continuously generates a virtual representation of the unknown remote environment. This virtual paradigm includes critical elements such as obstacles that may impede the robot's navigation, ensuring a comprehensive understanding of the surroundings. No map or global environmental model is provided in advance to the operator. As the robot explores and sends back new sensor data, the VR environment is updated in real time. This gives the operator an accurate and current view of only those areas that the robot has observed, supporting effective supervision

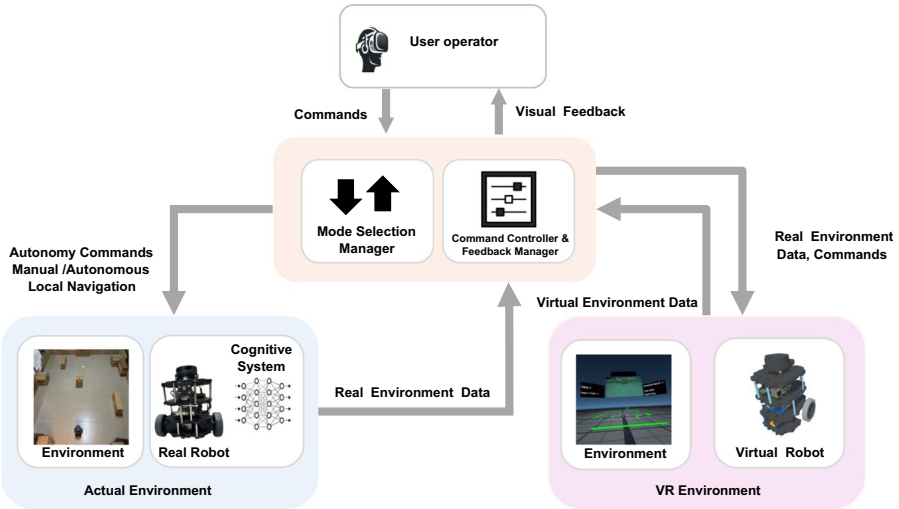


Fig. 1 Representation of the DSC concept, which integrates real and virtual environments. The VR environment generates an immersive understanding of the task environment for operators by utilizing data from a 2D-point cloud and live camera feed, paving the way for robots to effectively engage within their environments

and intervention.

- *Command Translation and Feedback*: Operator commands are seamlessly translated from the virtual environment to the actual robot, with real-time feedback provided to the operator. This feedback loop ensures that the operator remains informed of the robot's actions and can intervene if necessary.
- *Autonomous Navigation*: After the operator inputs the target position coordinates, the robot autonomously navigates to these points using a deep reinforcement learning (DRL) based cognitive system running on a local server. The operator monitors this process and can intervene as needed.
- *Mode Flexibility*: The operator has the flexibility to switch between manual operation and autonomous navigation, adjusting the distribution of high-level tasks (e.g., deciding destination waypoints) and low-level tasks (e.g., trajectory planning) based on the selected operational mode.

To optimize user experience and facilitate focus on critical elements, a dark sky theme is employed within the virtual environment. This thematic choice aids the operator in clearly discerning relevant features and obstacles that could impact on the robot's navigation. The operator is granted the ability to move around freely within the virtual space, though their physical movement is constrained by a VR boundary established at the outset of the operation. To overcome these physical limitations, omnidirectional treadmills can be integrated [55], enabling unrestricted movement within the VR space and allowing the operator to explore the virtual environment without physical boundaries. To enable intuitive and precise waypoint designation, the immersive VR environment integrates a virtual grid-based floor, structured into larger blocks corresponding precisely to 1×1-meter squares on the actual ground. Each large block is subdivided into four smaller squares measuring 0.5×0.5 m, enabling operator-friendly and accurate waypoint placement without the need for detailed global maps or sophisticated localization.

In addition to standard movement, the DSC system offers a teleportation feature that allows the operator to instantly move from one location to another within the virtual environment. By selecting a desired destination, the operator can teleport themselves to that point, with the option to adjust the direction of the "jump" as they make the selection. This feature enhances the operator's ability to quickly reposition within the virtual space, providing greater flexibility and efficiency in managing the robot's tasks.

By maneuvering the virtual twin, the operator effectively guides the real-world robot towards pre-established destinations. The DSC system operates under a distributed task scheme, wherein the operator sets waypoints that lead to the final goal as shown in Fig. 3. Once these waypoints are defined, the robot autonomously plans and follows a trajectory to reach them, ensuring that the overall task is completed efficiently and with minimal operator intervention. This autonomous capability is particularly valuable in dynamic and unpredictable environments, where real-time adjustments to the robot's path are necessary.

The comprehensive architecture of the DSC method is illustrated in Fig. 2. Within this system, interface units play a crucial role in processing operator commands and providing real-time feedback. These units ensure seamless communication between the operator and the robotic system, allowing for precise control and timely responses. The real and virtual robot units are responsible for executing the actions of both the physical robot and its virtual twin, maintaining synchronization between the two environments. This coordinated opera-

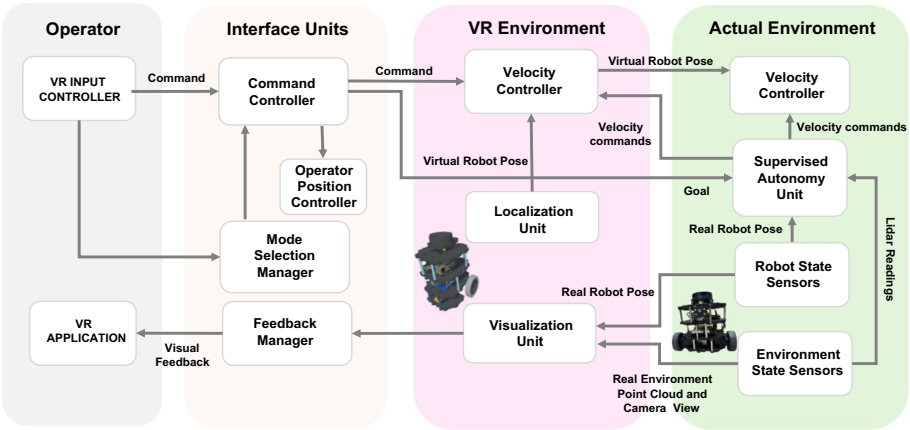


Fig. 2 Comprehensive system architecture of the proposed DSC teleoperation framework, illustrating the interaction and data flow between the operator, interface units, Virtual Reality (VR) environment, and the actual environment. The VR environment visualizes an unknown environment in real time, sending only those areas of the remote environment that are sensed and transmitted by the robot during exploration, without using any pre-existing map or environmental model for operator display

tion ensures that the virtual twin accurately mirrors the actions of the real robot, providing the operator with a reliable and intuitive control interface. By integrating these features, the DSC system enhances the operator's ability to manage complex tasks in remote environments, reducing workload while increasing the efficiency and effectiveness of the teleoperation process. All the units of Fig. 2 have been explicitly explained in the next section, 3.2, separately.

3.2 System units

3.2.1 Interface units

The interface units of the Distributed Supervised Control (DSC) system are critical for facilitating seamless interaction between the operator and the robotic system. These units comprise the mode selection manager, command controller, operator positioning controller, and feedback manager.

- *Mode Selection Manager*: This component is responsible for determining the operational mode of the system. The mode selection manager decides whether the operator will extend their supervision to include local navigation or opt for fully autonomous local operation. This choice directly influences how the system processes and executes commands, with further details on local autonomy provided in Sect. 3.3.
- *Command Controller*: The command controller processes the operator's inputs and translates them into motion commands for the virtual robot. These commands are adjusted based on the selected mode of operation, ensuring that the virtual robot responds accurately to the operator's directives.
- *Operator Positioning Controller*: The operator positioning controller tracks the operator's location within the virtual environment. This tracking ensures that the operator's

movements and interactions within the virtual space are accurately reflected in the system, allowing for intuitive and responsive control.

- *Feedback Manager*: The feedback manager plays a crucial role in maintaining a real-time connection between the operator and the robot. It communicates detailed information about the real robot's motion trajectory, including linear and angular velocities, heading, and position. Simultaneously, it streams a live video feed from the robot's camera into the virtual environment, providing the operator with a comprehensive view of the robot's surroundings and its actions.

3.2.2 Virtual and real environment units

The DSC system relies on a coordinated operation between its virtual and real environment units, each playing a critical role in ensuring the system's functionality. The virtual environment unit consists of the following key components:

- The *velocity controller* is responsible for publishing linear υ (m/s) and angular ω (rad/s) velocity commands to the virtual twin, ensuring that the virtual twin mirrors the movements of the real robot.
- The *visualization unit* manages the display of information on floating panels within the virtual environment, as depicted in Fig. 3. These panels provide crucial data to the operator, including the real robot's pose, velocities, remaining distance to the goal, obstacle warnings, and notifications when the destination is reached. Additionally, the visualization unit allows the operator to activate or deactivate a live 2D point cloud feed, which

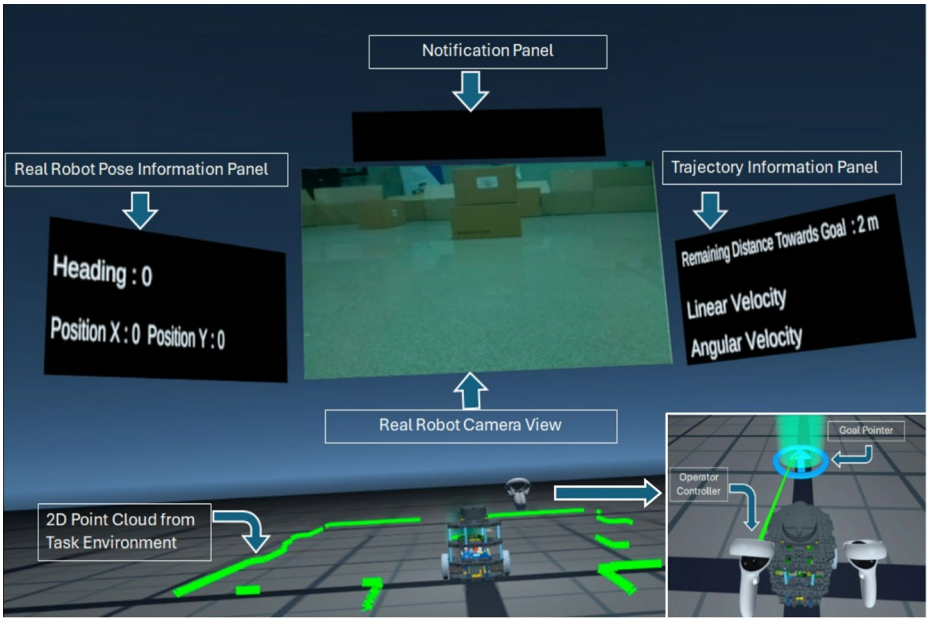


Fig. 3 Illustration of a VR application representing an unknown environment in a laboratory setting, highlighting the use of visualization units and the operator's interaction with the virtual environment. (**a**) Visualization unit (**b**) Operator providing navigation waypoints

overlays the virtual space floor to visualize object depth in the real environment. Real-world sensor data is streamed directly into the VR visualization unit via ROSBridge WebSocket, providing efficient topic-based communication, reduced latency, and seamless integration.
- The *localization unit* tracks the virtual robot's pose within the virtual reality (VR) environment, ensuring that the virtual twin's position remains consistent with that of the real robot.

Whereas real environment units include the following key components:

- The *environment* and *robot state sensor units* compile data from various sensors, including cameras and lidar, and publish this data as ROS (Robot Operating System) topics. The *environment state sensor unit* handles data related to the surroundings, while the *robot state sensor unit* gathers information on the robot's pose and movement from the wheel encoders.
- The *velocity controller* in the real robot operates similarly to its virtual counterpart, executing movement commands based on the inputs from either the supervised autonomy unit or the operator, ensuring precise control of the robot's actions.
- The *supervised autonomy unit* formulates a local trajectory using a goal-centric reinforcement learning policy upon receiving the operation goal coordinates. This policy enables the robot to autonomously navigate towards its goal while adapting to any obstacles or changes in the environment. If the operator chooses to take direct control, the autonomy unit uses odometry data from the virtual robot to calculate the necessary velocity commands, ensuring the real robot follows the intended path accurately.

These virtual and real environment units collectively enable the DSC system to function cohesively, providing the operator with the tools necessary to manage the robot's actions effectively across both virtual and physical domains. This architectural design demonstrates our proposed system in a remote, unknown/unmapped environment in a laboratory setting. In this context, as the mobile robot explores, its real-time sensor data (LiDAR point clouds and camera input) is streamed to the VR subsystem, which incrementally and dynamically updates the operator's VR perspective to reflect only those portions of the environment sensed by the robot at runtime. No pre-existing model, map, or prior environment knowledge is provided to the operator; the VR scene is constructed exclusively from the data acquired during the current exploration session of an unknown environment. Therefore, for the proposed experimental scenario, it is not a pre-built VR scenario; rather, the proposed solution recognizes it and provides a human user with the robot's interpretation of the explored environment for navigation in the real world.

3.3 Local autonomy

3.3.1 Navigation policy

To facilitate the actual robot, navigate autonomously after the goal position has been received from the operator, the DSC method employs a deep reinforcement learning based navigation policy. The navigation policy is devised using a goal-centric Twin Delayed Deep

Deterministic Policy Gradient (TD3) based reinforcement learning model [39, 56]. TD3 enables the execution of actions within a continuous action space through its actor-critic network. After locking in the goal coordinates, the robot's initial distance from and heading towards the goal point are calculated by (1) and (2), respectively. When the robot initiates movement, the relative angle between the heading of the robot and the direction to the goal is determined through computing angular disparity as per formula (3). This helps in obtaining the destination polar coordinates in the robot's frame of reference.

$$d_g = \sqrt{(\boldsymbol{P_{refx}} - \boldsymbol{P_{tx}})^2 + (\boldsymbol{P_{refy}} - \boldsymbol{P_{ty}})^2} \tag{1}$$

$$\alpha = tan^{-1} \frac{(\boldsymbol{P_{refy}} - \boldsymbol{P_{ty}})}{(\boldsymbol{P_{refx}} - \boldsymbol{P_{tx}})} \tag{2}$$

$$\theta_g = \alpha - \theta_t \tag{3}$$

The action-predicting (actor) neural network of TD3 is given a unique environment state vector (s), composed of laser scan readings, polar destination coordinates, and the previous action. It calculates an action from this vector, which is subsequently executed towards the destination point. The first network (actor neural network) comprises three fully connected (FC) layers, with Rectified Linear Unit (ReLU) activation applied after each layer while the final layer being linked to the output layer, featuring two action parameters *v*, and *w*, representing the robot's linear velocity and angular velocity, respectively. The output layer incorporates a hyperbolic tangent (tanh) activation function to constrain the values within the range of (−1, 1). Before application in the environment, the action a is represented as a tuple (*v*, *w*) is scaled by the upper bound of the linear velocity (v_{max}) and upper bound of the angular velocity (ω_{max}) using the following formula:

$$a = \left(v_{max} \left(\frac{v+1}{2} \right), w_{max}(w) \right) \tag{4}$$

The Q-value for a single state-action pair Q (s, a) is assessed by two critic networks. This helps with coping with Q-value overestimates. While both critic networks share the same structure, such as three fully connected layers, their parameter updates are staggered, allowing for divergence in parameter values. These networks take a pair of states and an action as input. State S undergoes processing through a fully connected layer followed by ReLU activation, resulting in the output O_s. This output, along with the action, is directed into two separate transformation fully connected layers (TFC) of identical size, κ_1 and κ_2, respectively. Subsequently, these layers are combined in the following manner:

$$O_c = O_s \omega_{\kappa_1} + a\omega_{\kappa_2} + B_{\kappa_1} \tag{5}$$

Here, O_c denotes the unified fully connected layer (UFC), where ω_{κ_1} and ω_{κ_2} represent the weights of κ_1 and κ_2, respectively. Additionally, B_{κ_1} signifies the bias of the layer κ_2. Subsequently, ReLU activation is applied to the combined layer, followed by a connection to the output with a single parameter representing the Q value. The minimum Q value

from both critic networks is then chosen as the ultimate critic output to mitigate overestimation of the state-action pair value. The comprehensive network architecture is depicted in Fig. 4. The policy is rewarded based on the function:

$$R(s_t, a_t) = \begin{cases} R_g, \ if \ d_t < \epsilon \\ R_c, \ if \ collision \\ v_t - |w_t|, \ otherwise \end{cases} \tag{6}$$

The reward (R) for the state-action pair (s_t, a_t) at timestep t depends on three conditions. If the distance to the goal at the current timestep (d_t) is below the threshold ϵ, a positive goal reward (R_g) is applied. In the case of a detected collision, a negative collision reward (R_c) is applied. If neither of these conditions is met, an immediate reward is applied, determined by the current linear velocity (v) and angular velocity (w). To steer the navigation policy toward the specified goal, a delayed attributed reward method is utilized, as outlined in the following calculation:

$$R_{t-1} = R(s_{i-1} a_{i-1}) + \frac{R_{t-1}}{i} \ \forall \ i = \{1, 2, 3, \ldots .k\} \tag{7}$$

Here, "k" represents the count of preceding steps during which rewards are adjusted. This implies that the positive goal reward diminishes progressively over the last "k" steps leading up to reaching a goal. The network, after rigorous training, acquired a comprehensive local navigation policy capable of reaching a local goal and adeptly steering clear of obstacles based directly on the laser inputs, as described by algorithm 1.

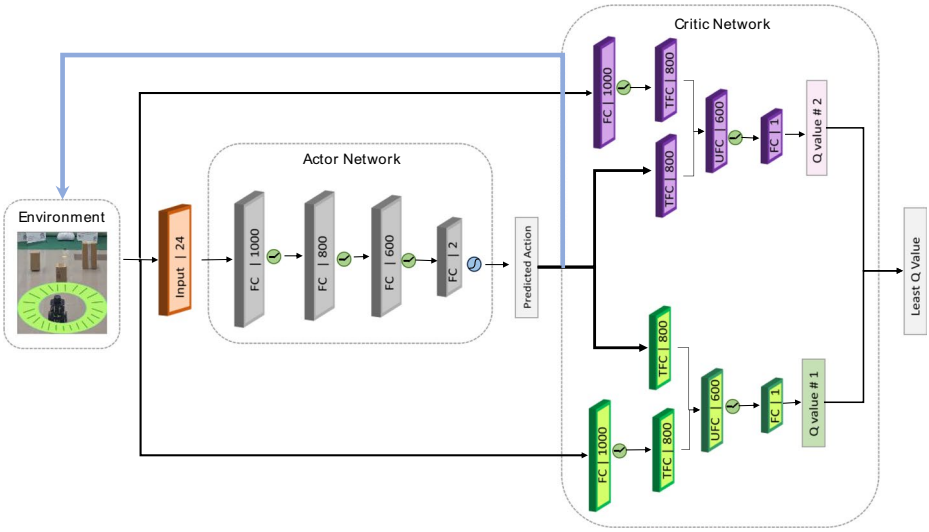


Fig. 4 This describes the deep reinforcement learning model network structure consisting of actor and critic parts

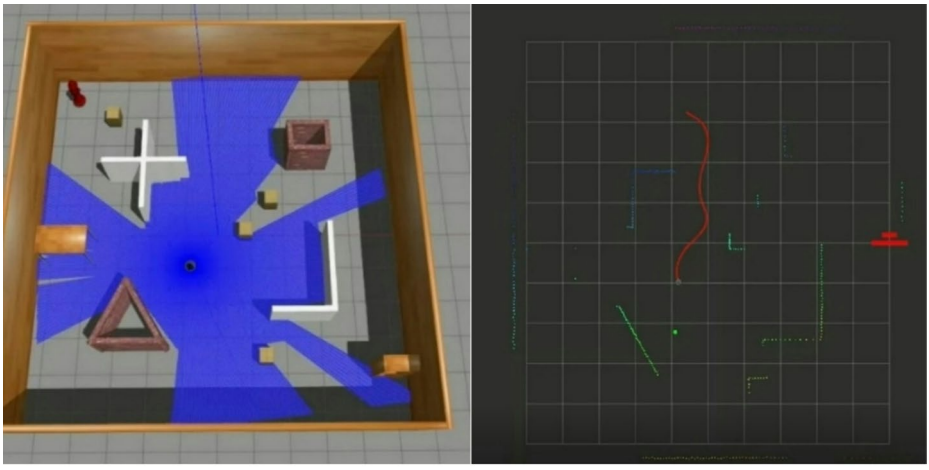
3.3.2 Training process

The DRL-based local navigation learning was conducted on a computer featuring an NVIDIA GTX 3080 graphics card, an Intel Core i7-10700 K CPU, and 32 GB of RAM. Training was executed in the physics simulator, Gazebo, for a total of 1200 episodes spanning roughly 10 h. Each training episode concluded upon achieving a target, identifying a collision, or completing 500 steps. Velocity limits $v_{\max}$ (maximum linear velocity) and $w_{\max}$ (maximum angular velocity) were set at 1.6 m per second and 1 rad per second, respectively. Delayed rewards were updated over the last $k = 12$ steps, with parameter updates occurring every 4 episodes. The training environment, a 12×12 m space cluttered with irregularly shaped obstacles depicted by Fig. 5, aimed to ensure the network's effectiveness in real-world scenarios by acquiring a generalized obstacle avoidance capability from lidar scan data. Gaussian noise was added to sensor and action values to foster generalization and policy exploration, while random adjustments to obstacle positions in each episode enhanced simulation-to-real transfer.

Incorporating the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm, the training used two Q-networks to mitigate overestimation bias and employed delayed policy updates for stability. Soft updates of target networks were governed by the equation:

$$\theta' = \tau\,\theta + (1 - \tau)\theta' \tag{8}$$

Where θ' represents the parameters of the target network, θ represents the parameters of the current network, and τ is a factor (typically set to a small value like 0.005) that determines the update rate. This equation ensures that the target network parameters are slowly adjusted towards the current network parameters, which helps in maintaining stability and avoiding large oscillations during training. These strategies, combined with environmental randomization, significantly improved the model's robustness and generalization, resulting in a network capable of effective obstacle avoidance from lidar data in real-world scenarios



(a) (b)

Fig. 5 Training overview. (**a**) The identical box obstacles are randomly placed after each episode. (**b**) The RVIZ highlights the robot's path while reaching the goal marker

4 Experimentation

In alignment with references [11, 30, 57, 58], a comprehensive evaluation of the proposed control system was conducted through both quantitative and qualitative user studies. The study involved a carefully selected group of approximately 19 participants, aged between 20 and 30, with varying levels of teleoperation experience. The number of participants was determined based on the referenced studies and further validated through a power analysis to ensure an adequate sample size for robust statistical analysis. Participants, comprising students and researchers from various engineering disciplines, were not exclusively focused on robotics, ensuring a diverse representation of teleoperation expertise. The study design and experimental procedures were rigorously reviewed and approved by the Ethics Review Committee of the School of Mechanical and Manufacturing Engineering at the National University of Sciences and Technology. This methodological approach ensured a thorough examination of the control system's effectiveness, leveraging both objective metrics and subjective feedback to provide a holistic assessment. The diverse background of participants allowed for a comprehensive evaluation, reflecting a broad spectrum of user experiences and perspectives.

4.1 Modes of operation

The assessment procedure utilized a 3×1 approach, where users were directed to select one mode of operation at a time from the provided list to complete the entire task. The modes of operation were as follows:

- Direct Control (DC): In this mode, the actual robot imitates its virtual twin, which is being directly operated by the user. The robot itself possesses no autonomy.
- Assistive Direct Control (ADC): This mode is like Direct Control but includes additional warning notifications about obstacles in proximity to the robot.
- Proposed Method: This method, elaborated in Sect. 3, incorporates advanced features designed to enhance the robot's performance and user interaction.

This structured approach allowed for a comprehensive evaluation of each control method, facilitating a detailed comparison of their effectiveness and user experience.

4.2 Task description

During the study, the participants were instructed to sit in a chair, put on a VR headset, and execute a rescue operation using the VR headset controllers. Participants were required to choose each mode of operation to assist the robot in navigating from the starting location to a predetermined destination, while circumventing obstacles. The task was to be executed once for each method without a pre-determined selection order to prevent user skill enhancement. Immediately after operating in a selected mode, participants were asked to fill out a workload questionnaire and system interface feedback questionnaires following the conclusion of the entire experiment.

4.3 Task hardware resource and operating environment

The mobile robot used for experimentation was a Turtlebot3 burger variant Fig. 6(c). It is a non-holonomic differential drive robot with a 0.08 m body radius. The robot was equipped with a Raspberry Pi camera module v2, positioned about 0.172 m above ground. The camera has a horizontal field of view of 62.2 degrees and a vertical field of view of 48.8 degrees. In all experiments, operator perception and control decisions were based exclusively on live data received from the robot as it explored a previously unmapped environment. The VR interface was continuously updated from this streaming sensor data and the robot's camera view for scene generation; no pre-existing maps or models were used for operator visualization and scene generation. The video feed was first compressed on the robot's onboard computer before it was transmitted to the Virtual Reality application. Sensor data, including compressed camera video streams and LiDAR-based point cloud information, were published by the robot as ROS topics, streamed to a local ROS master server, and then transmitted to the VR application using ROSBridge WebSocket interface over Wi-Fi. The VR application was run on an Oculus Quest 2 VR headset, featuring a 1832×1920-pixel display resolution per eye, a refresh rate of 90 Hz, 128 GB of storage, and 6GB of RAM. A Qualcomm Snapdragon XR2 processor powers it, making it capable of executing sophisticated VR Android applications. Along the navigation pathway, multiple obstructions approximately 0.2 m tall were arranged. The operator was privy to the two-dimensional point cloud depicted by the robot as well as the camera view and had the option to turn on the audio feed from the robot's microphone. The operational environment was established in a room separate from the operator's, effectively insulating the operator from perceiving the robots' sounds and movements. A TP Link TL-WR840N Wi-Fi router with a 300 Mbps transfer rate was utilized to connect the robot's onboard computer with the server at the operator end. An average latency of 0.25 s was noticed, which is fairly low and does not substantially affect the efficiency of a 2D.O.F system [59]. This latency was gauged by filming the participant's controller and mobile robot at a rate of 60 FPS and noting the delay period between the command instructions and the robot's movement. The experimentation involved directing the robot forward, backward, to the right, and to the left five times each, after which the average latency was calculated and reported. However, the latency for waypoints might change for complex real-world scenarios. Conversely, our proposed DSC method addresses this concern by reducing the sensitivity to latency by requiring the operator to provide only high-

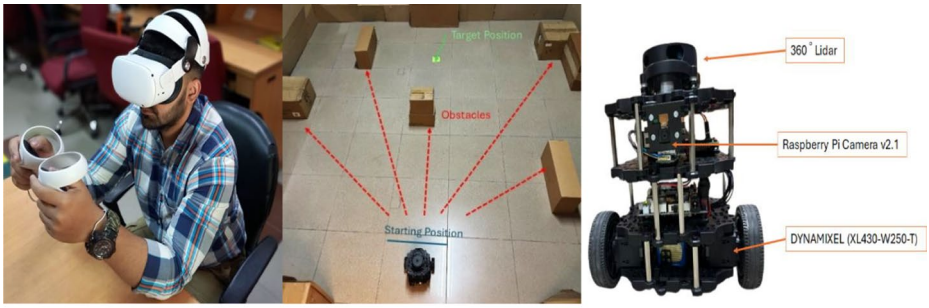


Fig. 6 Experimental Setup (**a**) Subject performing teleoperation task (**b**) Task course (**c**) Robot used for experimentation

level real-time navigational waypoints rather than continuous real-time corrections. Thus, moderate latency conditions do not significantly degrade the performance of our approach.

4.4 Hypothesis

The objective of the experiment was to ascertain whether the method was effective for (1) mitigating the teleoperation workload, (2) improving its task efficiency, and (3) providing better practicality with an immersive experience. In line with this aim, we formulated the following hypotheses:

- "H1: The DSC method ensured operation with decreased workload utilizing minimal information. ".
- "H2: The DSC method resulted in improved task efficiency. ".

In evaluating H1, the operation task load was assessed by analyzing the answers from a questionnaire. The participant's workload was evaluated utilizing the NASA-TLX [60] comprised of six domains: mental demand, physical demand, temporal demand, performance, effort, and frustration level. Each domain was assessed on a 10-point Likert scale, ranging from 0 to 10. We presumed that DSC would register a lower score relative to DC and ADC. In scenarios with limited visual information, the operator will have to repeatedly rethink their actions, causing task load to increase.

To validate H2, the efficiency of the task was assessed by comparing the following measures:

- Task execution time: This refers to the time duration required for the mobile robot to travel from the starting position to the destination.
- Collision frequency: This refers to the count of instances where the robot encountered collisions with obstacles.

In alignment with references [61–64] the system's interface usability and its quality are measured through various methodologies, including conventional application usability tests and user interviews, which were implemented to underscore the benefits of the suggested approach. Thereby, to access the operator feedback on the overall system usability and its immersivity, the participants in the study were asked to complete two established questionnaires: the Igroup Presence Questionnaire (IPQ) [65–67], and the System Usability Scale (SUS) [68] after completing a rescue operation in a remote arena settled for the robot was set to navigate in.

The immersive factor was assessed using the I group Presence Questionnaire (IPQ), comprising 13 questions with a likability scale of 1–7, the IPQ evaluates three key subscales:

- Spatial Presence: indicating the realization of the user's part of being physically present in the Virtual Environment.
- Engagement: measuring the attention and involvement experienced in the Virtual realm.
- Experienced Realism: assessing the subjective perception of realism within the Virtual Environment.

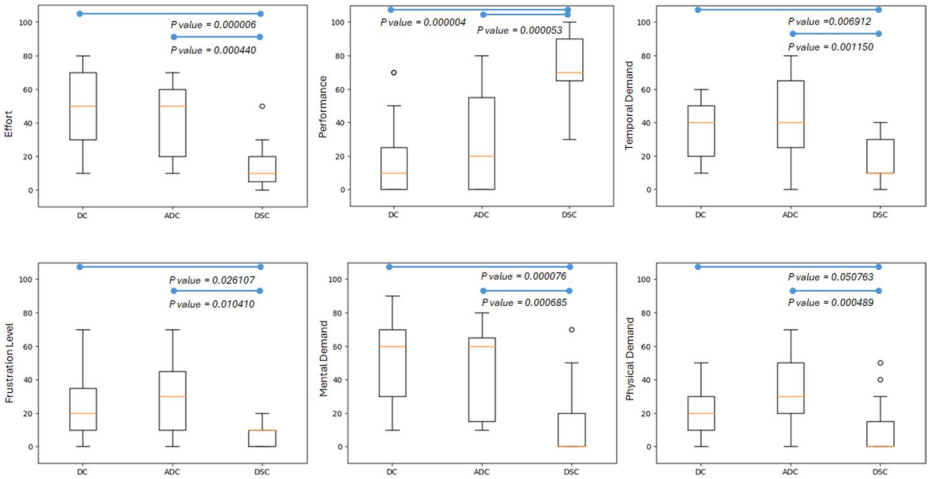


Fig. 7 Findings pertaining to Task load based on subjective evaluations. The comparative analysis between DSC and conventional methods was done using a follow-up Dunn's test

Table 2 Findings from task efficiency evaluations

Method	Task Execution Time[s]	Collision Frequency [Times]
DC	27.1 ± 6.38	0.2 ± 0.52
ADC	22.6 ± 2.34	0.15 ± 0.37
DSC	10.0 ± 0.2	0.0 ± 0.0

Additionally, the IPQ incorporates a general item gauging the overall "sense of being there," with notable emphasis on dimensional awareness. While developing an assessment of overall interface consistency in its usability, SUS was utilized, which consists of 10 questions, with five each for positive and negative impact review.

5 Results and deliberation

5.1 Reduced task load with limited information

The findings from the subjective workload evaluation are illustrated in Fig. 7. This data was initially screened through the Kolmogorov-Smirnov test to check for its normality. Since the data turned out not to be normally distributed. Non-parametric Kruskal-Wallis test was utilized to verify significant differences between group medians, followed by Dunn's test to identify which group is significantly different from the rest. The significance level was set at 0.05 for each test.

H1 stated that the DSC method would ensure an operation with a lower task load than DC and ADC. Significant differences were pointed out by the Kruskal-Wallis test for each domain of NASA-TLX; effort ($X^2 = 24.869$, P-value $= 3.98 \times 10^{-6}$), mental demand ($X^2 = 20.65$, P-value $= 3.28 \times 10^{-5}$), performance ($X^2 = 27.54$, P-value $= 1.05 \times 10^{-6}$), frustration ($X^2 = 9.932$, P-value $= 0.0069$), physical demand ($X^2 = 14.38$, P-value $= 0.00075$), and Temporal demand ($X^2 = 14.37$, P-value $= 0.00076$). Further,

Dunn's test highlighted how DSC significantly differed from DC and ADC in each domain, as illustrated on the Fig. 7. Thus, H1 was fully supported.

The reduction in overall task load demand can be attributed to the distribution of workload. With the DSC method, operators, regardless of their skill level, can rely on the robot's autonomy to avoid obstacles while navigating towards the final goal. This autonomy allows operators to focus on monitoring the robot's actions, rather than repeatedly reassessing and worrying about obstacles, which are necessary with DC and ADC methods. This shift in responsibility relieves operators from continuous cognitive and physical stress, thereby reducing the overall task load. By leveraging autonomous capabilities, the DSC method optimizes the task load distribution, enhancing operational efficiency and reducing the cognitive and physical demands on the operator. This supports the argument that advanced autonomous systems can significantly improve human-robot interaction by alleviating the workload and enhancing performance outcomes.

5.2 Improvement in task efficiency

H2 stated that the DSC method resulted in improved task efficiency. Table 2 displays outcomes from the task efficacy comparison about task execution duration and the frequency of obstacle collisions for all the methods. Here, DSC had the quickest time of execution, trailed by ADC and then DC. The reason is that the robust model training with accurate robot motion modelling enabled it to perform stable, relatively high-speed motion, resulting in less task execution time, while DC and ADC show a slower time as they suffer from the operator's skill to drive the robot. Figure 8 shows an example instance of a trajectory followed by the robot while being operated through DSC. Moreover, the DSC had the least number of collisions. However, the motion planning model's functionality can be further synergized with more environmental data through 3D lidar or a depth camera.

The task efficiency, measured by task execution time and collision frequency, was analyzed using Kruskal-Wallis's test, followed by Dunn's post-hoc test to explore specific differences between the methods. Given the nature of the dataset, the Kruskal-Wallis test was selected for its appropriateness in handling the data.

Algorithm 1 Goal-driven local autonomy

destination ► Obtain global Goal coordinates from VR operator
λ ► Set threshold for navigating to global goal
While (destinationReached == False) **Do**
  Obtain Robot Pose
  Read Sensor Data
  Update Environment State Vector (LDS readings, goal direction, $\mathbf{d}_t$)
  Update Robot State Vector (Environment State, Last action ($\mathbf{v_p}$, $\mathbf{w_p}$))
  if $d_t < \varepsilon$ **then**
    if waypoint = destination then
      destinationReached = True
    else if d ($\mathbf{p}_t$, $\mathbf{g}$) < λ **then**
      waypoint ← destination
    end if
  end if
Obtain **TD3_action***
Perform Action ($\mathbf{v_c}$, $\mathbf{w_c}$)
end while

*Action computed from Twin Delayed Deep Deterministic Model

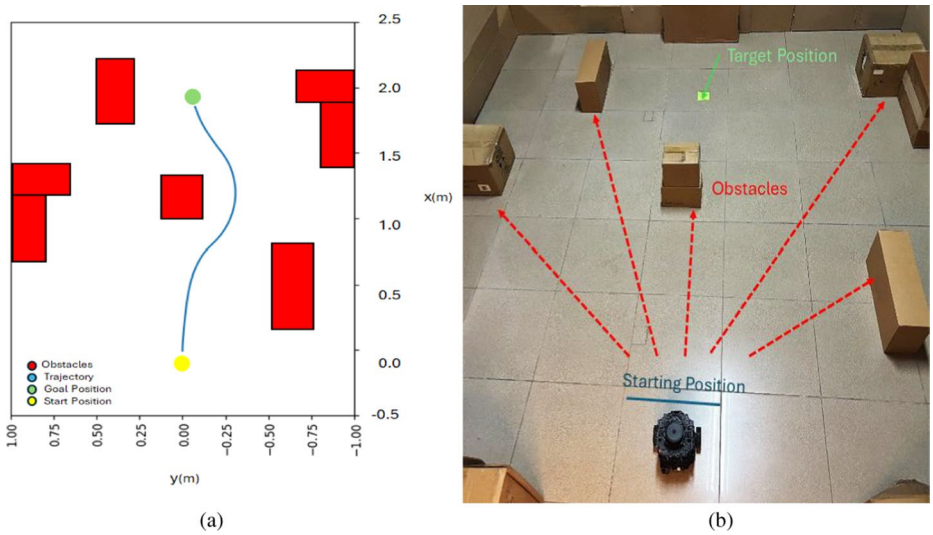


Fig. 8 (**a**) Trajectory planned by the Goal-centered TD3 network. (**b**) Task course

5.2.1 Task execution time

The Kruskal-Wallis test identified significant differences in task execution times across the three methods, with an H-statistic of 2125.78 and a p-value of 0.0, indicating variation in task execution times between the methods. To further investigate these differences, Dunn's post-hoc test was conducted. The results showed that the DSP method differed significantly from both DC and ADC:

- DC vs. DSP: The analysis revealed a significant difference ($p \approx 4.31\text{e-}305$), with DSP showing a much lower execution time.
- ADC vs. DSP: A significant difference was also observed ($p \approx 4.70\text{e-}284$), with DSP outperforming ADC in terms of execution time.
- DC vs. ADC: The comparison between DC and ADC did not show a significant difference ($p = 0.562$), suggesting similar execution times for these two methods.

These results suggest that DSP consistently outperformed the other methods in task execution time. In contrast, DC and ADC did not differ significantly in their execution times. The visual representation through violin plots illustrated in Fig. 9a supported these findings. The DSP method exhibited the lowest task execution time with minimal variance, indicating efficient and consistent performance. The DC method, however, showed higher variance, suggesting less consistency. The violin plots further illustrate the tight distribution around DSP's lower execution times, while DC and ADC displayed broader distributions.

5.2.2 Collision frequency

The Kruskal-Wallis test for collision frequency also indicated significant differences among the methods, with an H-statistic of 189.94 and a p-value of 5.6878e-42. This suggests that

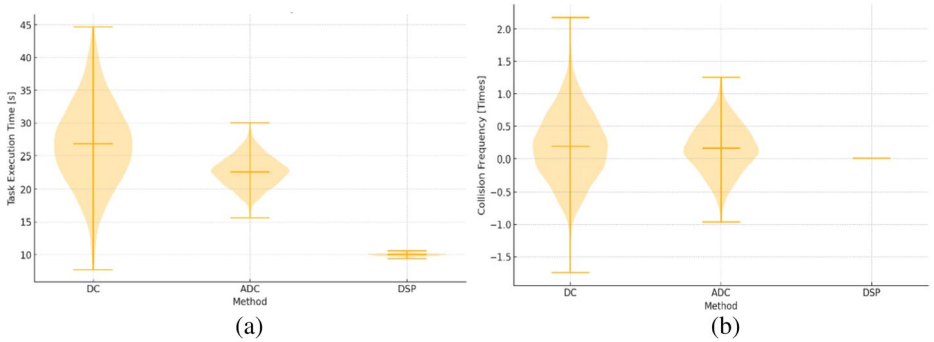


Fig. 9 (**a**) Collision frequency by method. (**b**) Task Execution Time by Method

the methods varied in their effectiveness at minimizing collisions. Dunn's post-hoc test results for collision frequency provided additional clarity:

- DC vs. DSP: A significant difference was observed ($p \approx 2.59\text{e-}30$), with DSP demonstrating no collisions.
- ADC vs. DSP: Similarly, a significant difference was found ($p \approx 2.79\text{e-}34$), with DSP again showing no collisions.
- DC vs. ADC: The test did not reveal a significant difference between DC and ADC ($p = 1.000$), indicating comparable collision frequencies for these two methods.

These findings suggest that DSP was notably more effective at avoiding collisions compared to DC and ADC, while DC and ADC showed similar collision frequencies. The visual analysis, violin plots illustrated in Fig. 9b, reinforced these observations. The DSP method was associated with zero collisions, reflecting its effectiveness in obstacle avoidance. In contrast, the DC method showed greater variance in collision frequency, indicating less consistent performance in this regard. Dunn's post-hoc test results highlight that the DSP method exhibited significant improvements in both task execution time and collision frequency compared to DC and ADC. DSP consistently demonstrated lower execution times and zero collision frequency, which may suggest its potential as an efficient method for tasks requiring precision. The lack of significant differences between DC and ADC in both measures indicates that these methods perform similarly, although neither matches the efficiency observed with DSP.

5.3 Power analysis

To enhance the reliability and robustness of our study's findings, we conducted a thorough post-hoc power analysis to calculate the actual statistical power achieved. This analysis, fundamental for verifying the study's adequate power to detect the observed effect size, revealed insightful outcomes. Initially, the effect size was determined using the Kruskal-Wallis test, indicating that a sample size of approximately 17 participants per group would be required to attain a desired power of 0.80, meeting standard empirical research thresholds as illustrated by Fig. 10. However, our study included 19 participants per group, exceeding the minimum requisite sample size. This increase in sample size significantly boosted the

statistical power to 0.84, reflecting a high probability of correctly detecting the observed effect size if present in the population. Such elevated power enhances confidence in our findings and mitigates the risk of Type II errors—situations where a false null hypothesis might be mistakenly accepted, thereby failing to detect a true effect. Consequently, this robust power level substantiates the reliability of our results, reducing the occurrence of Type II errors and ensuring that the observed outcomes are not erroneously dismissed. This credibility provides a solid foundation for subsequent research and practical applications based on our findings.

5.4 Interface usability

The System Usability Scale (SUS) questionnaire results revealed that the assessed usability of the system achieved a commendable score of 84.63 out of 100, with a range spanning from a minimum score of 79.5 to a maximum of 100, and a standard deviation of 5.34. These results indicate that the presented immersive framework interface attained a significantly high level of usability. Figure 11(b) provides a visual representation of the mean scores and standard deviations for each question in the SUS questionnaire. Notably, a majority of participants expressed a strong willingness to frequently use this interface, primarily due to its user-friendly nature. The interface was reported to be easy to navigate, with users particularly appreciating the seamless integration of all functionalities. Additionally, the interface was perceived as consistent, instilling a high level of confidence in its usability among the users.

The Immersive Presence Questionnaire (IPQ) was also utilized, incorporating a general item that gauges the overall "sense of being there," with particular emphasis on dimensional awareness. The feedback from the IPQ is depicted in Fig. 12. Specifically, Fig. 12(a) presents the average user ratings along with their dispersion for each IPQ question, while

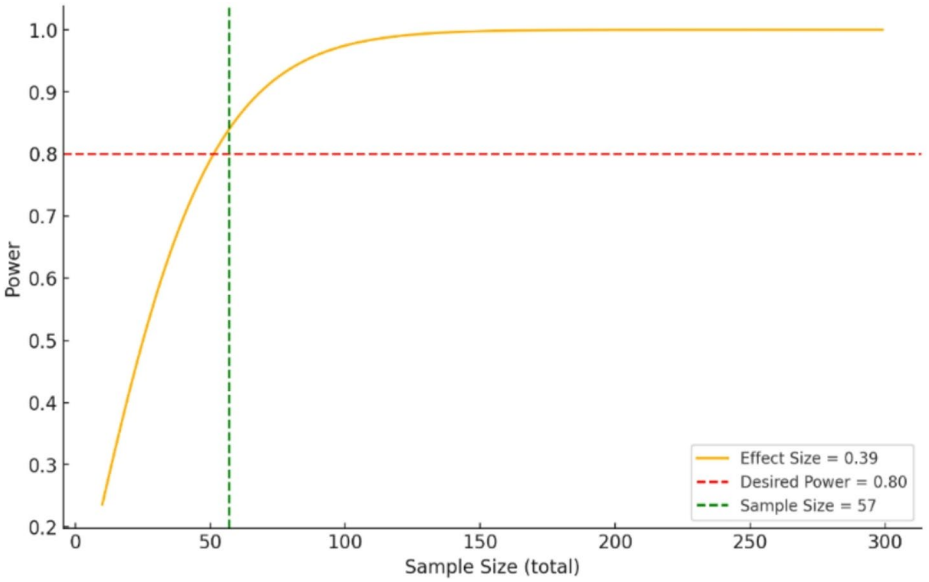
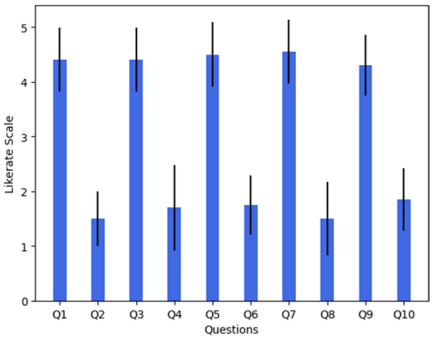


Fig. 10 (**a**) Power analysis for Kruskal-Wallis test

Id	Statement	Score
1	I think that I would like to use this system frequently?	4.368 ± 0.598
2	I found the system unnecessarily complex?	1.474 ± 0.513
3	I thought the system was easy to use?	4.421 ± 0.598
4	I think that I would need the support of a technical person to be able to use this system?	1.684 ± 0.801
5	I found the various functions in this system were well integrated?	4.526 ± 0.607
6	I thought there was too much inconsistency in this system?	1.736 ± 0.550
7	I would imagine that most people would learn to use this system very quickly?	4.526 ± 0.605
8	I found the system very cumbersome (slow or complicated) to use?	1.527 ± 0.688
9	I felt very confident using the system?	4.263 ± 0.571
10	I needed to learn a lot of things before I could get going with this system?	1.842 ± 0.587
Overall Usability Score		84.625 ± 5.336

(a)

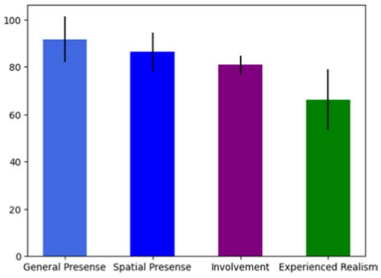


(b)

Fig. 11 Results accumulated after user feedback on System Usability Scale Questionnaire (**a**) Overall Usability Score Calculation (**b**) Mean score for each question along with deviation computed

Id	Statement	Score
1	In the computer-generated world I had a sense of "being there"?	6.421 ± 0.692
2	Somehow, I felt that the virtual world surrounded me?	6.053 ± 1.079
3	I felt like I was just perceiving pictures?	2.211 ± 1.032
4	I did not feel present in the virtual space?	1.737 ± 0.562
5	I had a sense of acting in the virtual space, rather than operating something from outside?	5.842 ± 0.958
6	I felt present in the virtual space?	5.789 ± 0.787
7	How aware were you of the real world surrounding while navigating in the virtual world? (i.e., sounds, room temperature, and other people)?	1.158 ± 0.501
8	I was not aware of my real environment?	6.789 ± 0.535
9	I still paid attention to the real environment?	1.263 ± 0.561
10	I was completely captivated by the virtual world?	6.737 ± 0.653
11	How real did the virtual world seem to you?	5.474 ± 1.073
12	How much did your experience in the virtual environment seem consistent with your real-world experience?	5.053 ± 1.544
13	The virtual world seemed more realistic than the real world	3.475 ± 1.020

(a)



(b)

Fig. 12 Results accumulated after user feedback on igroup presence questionnaire (**a**) Mean score for each question along with deviation computed (**b**) Scores calculated for each subdomain

Fig. 12(b) shows the total percentage for each IPQ subscale. The general presence score was exceptionally high, reaching 95.45% with a standard deviation of 2.2, indicating that users felt a strong sense of being inside the virtual environment (VE). The spatial presence score was even higher, at 99.5% with a standard deviation of 2.8, suggesting that users felt as though they were physically present within the VE. The involvement score, recorded at 94.1% with a standard deviation of 1.6, supported the findings from the SUS questionnaire, confirming that users were actively engaged and maintained focus on all elements within the VE.

On the other hand, the realism rating encountered by users registered at 44.3% with a standard deviation of 16.2, signifying that while users consistently perceived themselves as being within a VE, the environment itself was not characterized by high realism. This

outcome aligns with the intended goal of the proposed approach, which prioritized creating a natural and user-friendly VE over achieving heightened realism. This design choice was made with the consideration of the computational costs associated with rendering highly realistic environments, as well as the need for specialized hardware, such as high-performance GPUs, which may not be accessible to all users.

Using the System Usability Questionnaire and the Igroup Presence Questionnaire, which measures usability and factors including general presence, spatial presence, involvement, and experience realism, indicates the reason for the use of VR as a tool to enhance immersivity and user experience.

The use of Virtual Reality (VR) as a tool to enhance immersivity and user experience is supported by the results from the System Usability Questionnaire and the Igroup Presence Questionnaire, which assess usability along with key factors such as general presence, spatial presence, involvement, and perceived realism of the experience.

6 Conclusion

The proposed method introduces a novel teleoperation technique termed Distributed Supervisory Control (DSC), aimed at (1) reducing teleoperation workload, (2) improving task efficiency, and (3) enhancing overall system suitability, including providing a more immersive user experience while reducing computational overhead. The DSC approach accomplishes these objectives by integrating task division, parallel supervisory control, and a goal-oriented twin delayed deep deterministic policy gradient (TD3) agent. This configuration enables human operators to concentrate on higher-order tasks, such as setting destination waypoints, while autonomous robotic systems manage lower-order tasks like obstacle avoidance and trajectory plotting with minimal state information. The incorporation of visual feedback coupled with parallel supervisory control enhances user perceptual understanding while minimizing computational burden and processing costs. Additionally, onboard data compression and efficient messaging protocols significantly mitigate data volume, latency, and computational load. The evaluation study validated the method's effectiveness and practicality through real-time experimentation using metrics including NASA-TLX, task execution time, and obstacle collision frequency. User studies further demonstrated enhanced user experience, assessed via the System Usability Scale (SUS) and Igroup Presence (IPQ) questionnaires for immersive experience. A power analysis was conducted to determine the appropriate number of participants for the study, ensuring that the sample size was statistically adequate to validate the findings. Statistical analysis revealed significant performance improvements with our DSC method relative to Direct Control (DC) and Assistive Direct Control (ADC), thereby enhancing task efficiency and overall interface suitability. In conclusion, the DSC method establishes a new paradigm for remote operations in diverse and dynamic environments. The DSC approach is especially applicable for real-world scenarios involving exploration of unknown or hazardous environments such as disaster response missions, remote inspection of industrial facilities, and search-and-rescue operations.

7 Future work

Several promising avenues for future research naturally arise from this study. Firstly, extending the Distributed Supervisory Control (DSC) method explicitly to multi-robot teleoperation represents an essential step towards broader robotic operational scalability. Secondly, additional robustness measures should be explicitly investigated to handle unforeseen failures or malfunctions in critical robot subsystems, such as perception, localization, planning, and control modules. Moreover, future evaluations should explicitly explore and validate our method under varied and realistic latency conditions, including scenarios requiring quick real-time interactions. Moreover, future efforts could further investigate rapid real-time 3D reconstruction of unstructured environments with minimal communication lag using immersive VR, significantly enhancing immersion and operator effectiveness.

Author contributions Conceptualization, F.M. and S.A.; methodology, F.M., S.A., M.A.K; software, F.M., Y.A., S.A.,; validation, F.M., K.J. and M.B.S.; formal analysis, F.M.; investigation, M.B.S.; resources, M.A.K., Y.N; data curation, S.A., F.M.; writing—original draft preparation, F.M.; writing—review and editing, S.A., M.A.K., Y.N.,; visualization, M.B.S., S.A.; supervision, S.A., K.J., Y.A.,; project administration, S.A.; funding acquisition, M.A.K., Y.N.

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Data availability The data presented in this study is available on request to the corresponding author.

Declarations

Ethical approval The Ethical Review Committee of the National University of Sciences and Technology (NUST), i.e., the NUST institutional Review Board has approved the study and issued the IRB certificate to conduct experimentation.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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